

Midterm Review

Estimability • model updating • MVN & χ^2 • quadratic forms • Cochran • hypothesis testing

$C(X)$

estimability

MVN

χ^2

Cochran

F

What the midterm should test

Focus on connections among algebra, geometry, and probability.

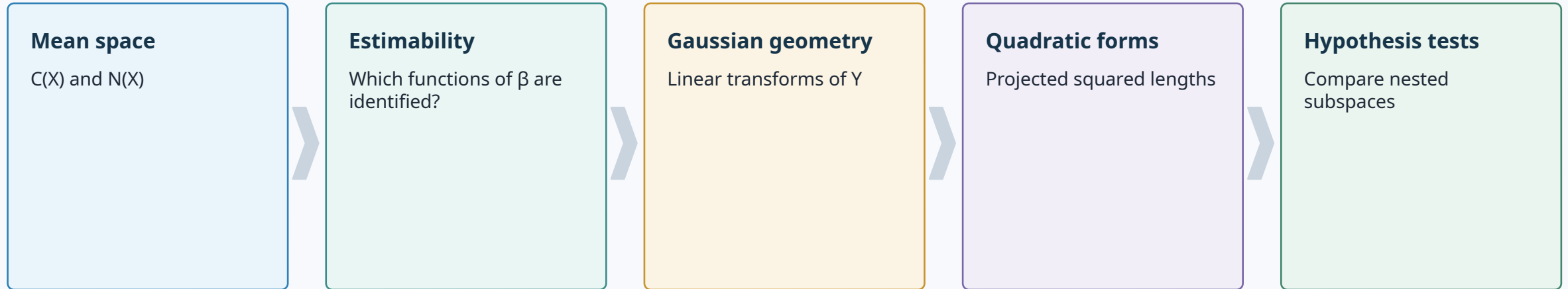
SCOPE

- Determine what is estimable or testable in full- and less-than-full-rank models.
- Use low-rank updates to understand what changes when a covariate or observation is added or removed.
- Manipulate multivariate normal vectors and identify when Gaussian components are independent.
- Recognize when quadratic forms are central or noncentral chi-squared.
- Use Cochran's theorem to justify independence of sums of squares.
- Construct and interpret t and F tests for linear hypotheses.

The midterm in one chain

Each topic feeds the next.

BIG PICTURE



Geometry tells us what can be estimated; normal theory tells us its exact distribution.

Which theorem should I reach for?

A fast decision map for exam problems.

STRATEGY

Question	Tool
Is $c^T\beta$ identifiable?	$c \in C(X^T)$, or $c \perp N(X)$
How does a fit change after adding z ?	Residualize z ; projection update / FWL
Distribution of $AY+b$?	MVN transformation rule
Distribution of Y^TPY ?	Projection quadratic-form theorem
Are two sums of squares independent?	Orthogonality + Cochran's theorem
Test $C\beta=d$?	General linear F test

01

Estimability

Rank deficiency changes parameter uniqueness, but not every inferential target is lost.

Rank deficiency creates equivalent parameter vectors

Different β values can generate exactly the same mean vector.

RANK DEFICIENT

$$Xh = 0, h \neq 0 \quad \Rightarrow \quad X(\beta + h) = X\beta$$

Parameter space

β and $\beta+h$ are different points.

Mean space

Both map to the same $X\beta$.

Inference

Only functions constant across the equivalence class can be estimated.

The estimability criterion

Three equivalent ways to recognize an estimable linear function.

CORE RESULT

$$c^T \beta \text{ estimable} \iff c \in \mathcal{C}(X^T) \iff c \perp \mathcal{N}(X)$$

Definition

There exists $a^T Y$ with $E(a^T Y) = c^T \beta$ for every β .

Row-space view

c must be a linear combination of rows of X .

Null-space view

$c^T h = 0$ for every unidentifiable direction h .

Worked example: a two-group effects model

Individual effects are not identified, but group means and contrasts are.

EXAMPLE

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\beta = (\mu, \alpha_1, \alpha_2)^T \quad h = (1, -1, -1)^T \in \mathcal{N}(X)$$

Not estimable

μ , α_1 , and α_2 individually.

Estimable

$\mu + \alpha_1$ and $\mu + \alpha_2$.

Estimable contrast

$\alpha_1 - \alpha_2$ because $(0, 1, -1)^T h = 0$.

Generalized inverses: nonunique coefficients, unique fit

A generalized inverse chooses a representative solution to the normal equations.

G-INVERSE

$$B \text{ g-inverse of } X^T X : \quad (X^T X)B(X^T X) = X^T X$$

$$\hat{\beta}_B = BX^T Y, \quad \hat{Y} = XBX^T Y$$

Can depend on B

The coordinates of $\hat{\beta}$.

Does not depend on B

$X\hat{\beta}$ and any estimable $c^T\hat{\beta}$.

Testability is estimability in disguise

A hypothesis is meaningful only if it concerns identified functions.

[LINK TO TESTING](#)

$$H_0 : C\beta = d$$

Rows of $C \subseteq \mathcal{C}(X^T) \implies C\beta$ is testable

Before computing an F statistic in a rank-deficient model, check that the restriction is estimable.

Checkpoint: estimability

Try to answer before looking back at the formulas.

CONCEPT CHECK

1

What does $Xh=0$ tell you about the uniqueness of β ?

2

Give two equivalent criteria for $c^T\beta$ to be estimable.

3

Why is $X\hat{\beta}$ unique even when $\hat{\beta}$ is not?

4

Why can an identifying constraint change $\hat{\beta}$ but not an estimable contrast?

02

Model updating

Low-rank algebra reveals what changes when the design matrix changes.

Adding a covariate: remove what X already explains

Residualization isolates the genuinely new direction.

ADD COVARIATE

$$M_X = I - H_X, \quad z^* = M_X z$$

If $z^* = 0$

z already lies in $C(X)$; the model space does not grow.

If $z^* \neq 0$

z^* is orthogonal to $C(X)$ and adds one new direction.

Geometric view

The enlarged model is $C(X) \oplus \text{span}(z^*)$.

Frisch-Waugh-Lovell theorem

The new coefficient depends only on the residualized response and residualized covariate.

FWL

$$\hat{\gamma} = \frac{z^T M_X Y}{z^T M_X z} = \frac{z^{*T} Y}{z^{*T} z^*}$$

Step 1

Regress Y on X and retain residuals.

Step 2

Regress z on X and retain z^* .

Step 3

Regress residualized Y on z^* .

Projection update and extra sum of squares

Adding one linearly independent predictor is a rank-one projection update.

RANK ONE

$$H_+ = H_X + \frac{z^* z^{*T}}{z^{*T} z^*}$$

$$SSE_X - SSE_+ = \frac{(z^{*T} Y)^2}{z^{*T} z^*}$$

Extra SS = squared length of Y projected onto the new orthogonal direction.

Adding one observation: Sherman-Morrison

A new subject changes $X^T X$ by a rank-one term.

ADD SUBJECT

$$X_+^T X_+ = X^T X + x_0 x_0^T$$

$$(X_+^T X_+)^{-1} = A - \frac{A x_0 x_0^T A}{1 + x_0^T A x_0}, \quad A = (X^T X)^{-1}$$

$$h_0 = x_0^T A x_0$$

Measures how far the new design point extends beyond existing information.

Rank-one update

No need to invert a new $p \times p$ matrix from scratch.

Deleting one observation

Residual size and leverage jointly determine the coefficient change.

DELETE SUBJECT

$$\hat{\beta}_{(i)} = \hat{\beta} - \frac{(X^T X)^{-1} x_i e_i}{1 - h_{ii}}$$

Large $|e_i|$

Observation disagrees strongly with its fitted value.

Large h_{ii}

Observation has unusual predictor coordinates.

Both large

Potentially large change in $\hat{\beta}$.

Leave-one-out identities

One fit gives all deleted residuals.

LOO

$$e_{(i)} = \frac{e_i}{1 - h_{ii}}$$

$$PRESS = \sum_{i=1}^n \left(\frac{e_i}{1 - h_{ii}} \right)^2$$

Interpretation

High leverage amplifies the gap between ordinary and deleted residuals.

Computational idea

Cross-validation can often be updated algebraically rather than refitting n times.

Checkpoint: model updating

Try to answer before looking back at the formulas.

CONCEPT CHECK

1

Why do we residualize z before asking what it contributes?

2

What does $z^*=0$ imply about the enlarged model?

3

What is the geometric meaning of the extra sum of squares from adding z ?

4

Why does leverage appear in deletion formulas?

03

Multivariate normal and chi-squared distributions

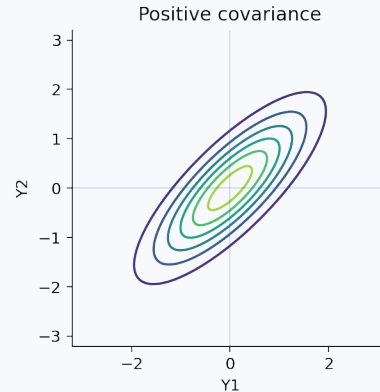
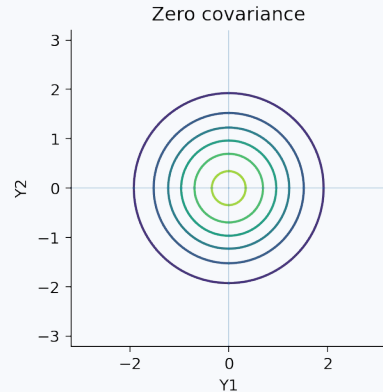
These are the probability tools that make exact linear-model inference possible.

Multivariate normal transformations

Any fixed linear transformation of an MVN vector is again MVN.

MVN

$$Y \sim N_n(\mu, \Sigma) \quad \Rightarrow \quad AY + b \sim N(A\mu + b, A\Sigma A^T)$$



Marginals

Select rows of A.

Contrasts

Choose A to form desired linear combinations.

For Gaussian vectors, zero covariance is enough

This special property drives many independence arguments in linear models.

INDEPENDENCE

$$\begin{pmatrix} U \\ V \end{pmatrix} \text{ jointly normal, } \quad \text{Cov}(U, V) = 0 \quad \implies \quad U \perp V$$

$$\text{Cov}(AY, BY) = A\Sigma B^T$$

In the normal linear model, prove independence by proving joint normality + zero cross-covariance.

Key Gaussian distributions in the linear model

Know these cold in the full-rank case.

LINEAR MODEL

$$Y \sim N_n(X\beta, \sigma^2 I)$$

$$\hat{\beta} \sim N_p(\beta, \sigma^2 (X^T X)^{-1})$$

$$\hat{Y} = HY \sim N_n(X\beta, \sigma^2 H)$$

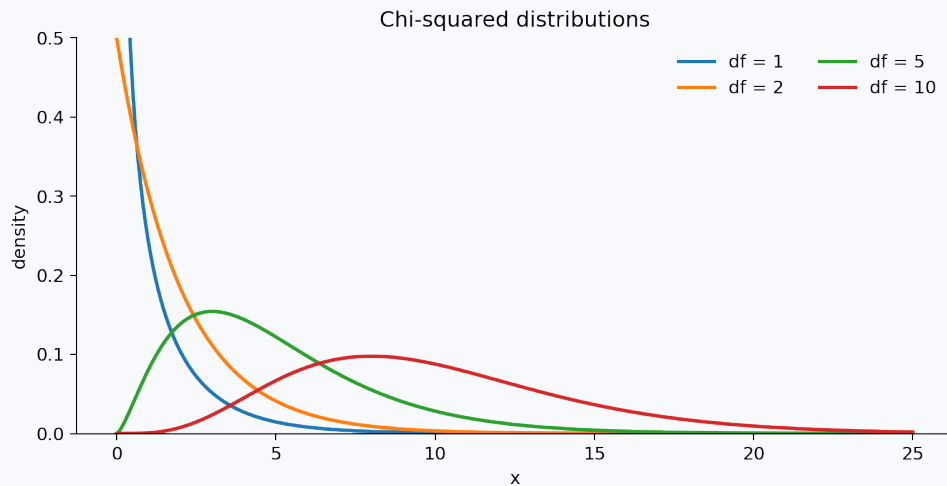
$$e = MY \sim N_n(0, \sigma^2 M)$$

$\hat{\beta} \perp e$ because $X^T M = 0$.

Chi-squared distribution

A chi-squared variable is a squared Euclidean length of a standard normal vector.

CHI SQUARED



$$Z_1, \dots, Z_r \stackrel{iid}{\sim} N(0, 1) \Rightarrow \sum_{j=1}^r Z_j^2 \sim \chi_r^2$$

Mean

$$E(Q) = r$$

Variance

$$\text{Var}(Q) = 2r$$

Degrees of freedom = number of independent squared normal coordinates.

Classical prototype: sample mean and sample variance

The same geometry reappears in regression.

PROTOTYPE

$$J = \frac{1}{n} \mathbf{1}\mathbf{1}^T, \quad M = I - J$$

$$\frac{Y^T M Y}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

Mean component

JY lies in $\text{span}(\mathbf{1})$, dimension 1.

Deviation component

MY lies in $\text{span}(\mathbf{1})^\perp$, dimension $n-1$.

Orthogonal Gaussian components are independent: $\bar{Y} \perp S^2$.

Regression residual sum of squares is the same idea

Replace the mean space $\text{span}(1)$ by the p -dimensional model space $C(X)$.

BRIDGE

$$M = I - H, \quad \text{rank}(M) = n - p$$

$$SSE = Y^T M Y = \varepsilon^T M \varepsilon$$

$$\frac{SSE}{\sigma^2} \sim \chi_{n-p}^2$$

Residual degrees of freedom are a dimension: $\dim C(X)^\perp = n - p$.

Checkpoint: MVN and χ^2

Try to answer before looking back at the formulas.

CONCEPT CHECK

1

How do you find the distribution of $AY+b$ from the distribution of Y ?

2

What extra fact does joint normality give you when a covariance is zero?

3

Why are regression residuals correlated even when the original errors are independent?

4

Why is $n-p$ the residual chi-squared degrees of freedom?

04

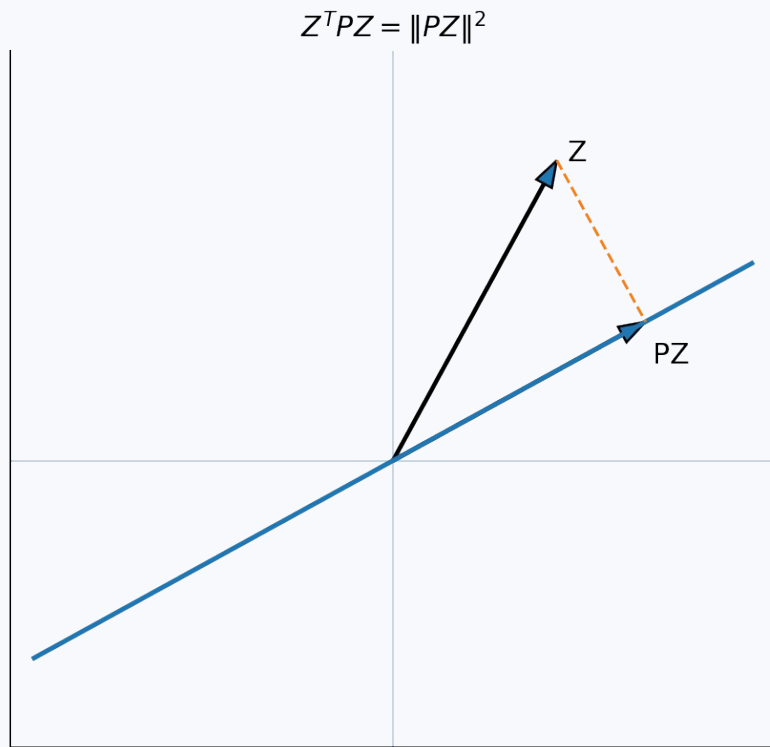
Quadratic forms and Cochran's theorem

Projection matrices turn normal vectors into independent sums of squares.

General normal quadratic forms

Spectral decomposition exposes the distribution.

QUADRATIC FORMS



$$A = Q\Lambda Q^T, \quad U = Q^T Z$$

$$Z^T A Z = \sum_{j=1}^n \lambda_j U_j^2$$

Key distinction

A general symmetric A gives a weighted sum of χ^2_1 variables—not usually a χ^2 distribution.

Projection quadratic forms are special

Symmetric idempotence forces eigenvalues to be 0 or 1.

PROJECTION

$$P = P^T = P^2, \quad r = \text{rank}(P)$$

$$Z \sim N(0, I) \quad \Rightarrow \quad Z^T P Z \sim \chi_r^2$$

$$Y \sim N(\mu, \sigma^2 I) \quad \Rightarrow \quad \frac{Y^T P Y}{\sigma^2} \sim \chi_r^2(\lambda), \quad \lambda = \frac{\mu^T P \mu}{\sigma^2}$$

Central iff $P\mu = 0$.

Cochran's theorem

Orthogonal projection pieces produce independent chi-squared variables.

THEOREM

$$P_j = P_j^T = P_j^2, \quad P_i P_j = 0 \ (i \neq j), \quad \sum_{j=1}^k P_j = I$$

$$Z^T P_j Z \sim \chi_{r_j}^2 \text{ independently,} \quad r_j = \text{rank}(P_j)$$

Distribution

Each piece is a projection quadratic form.

Independence

Different pieces use orthogonal Gaussian coordinates.

Rank accounting

The ranks add to n.

Regression sums of squares are orthogonal projections

With an intercept, corrected total splits into regression + error.

REGRESSION

$$I - J = (H - J) + (I - H)$$

$$SST = SSR + SSE$$

Corrected total

$P = I - J$, rank $n - 1$.

Regression

$P = H - J$, rank $p - 1$.

Error

$P = I - H$, rank $n - p$.

$(H - J)(I - H) = 0$, so $SSR \perp SSE$ under normality.

Nested models: the extra sum of squares

The difference between two nested projections is another projection.

NESTED MODELS

$$\mathcal{C}(X_0) \subset \mathcal{C}(X_1), \quad P_{extra} = H_1 - H_0$$

$$SSE_0 - SSE_1 = Y^T (H_1 - H_0) Y$$

Rank

$$\text{rank}(H_1 - H_0) = p_1 - p_0.$$

Orthogonality

$$(H_1 - H_0)(I - H_1) = 0.$$

Degrees of freedom are ranks, not memorized labels

Every df count is the dimension of a subspace.

DEGREES OF FREEDOM

Quantity	Projection	df
SSE	$I - H$	$n - p$
SSR with intercept	$H - J$	$p - 1$
Extra SS	$H_1 - H_0$	$p_1 - p_0$
Corrected SST	$I - J$	$n - 1$

When in doubt, find the projection matrix and take its rank.

Checkpoint: quadratic forms and Cochran

Try to answer before looking back at the formulas.

CONCEPT CHECK

- 1 Why is a symmetric quadratic form usually not chi-squared?
- 2 What property of a projection forces the eigenvalues to be 0 and 1?
- 3 What additional conclusion does Cochran's theorem give beyond the marginal χ^2 distributions?
- 4 Why is $H_1 - H_0$ a projection when the models are nested?

05

Hypothesis testing for linear models

Testing asks whether the mean needs directions excluded by the null model.

General linear hypothesis

Most classical tests can be written as $H_0: C\beta = d$.

GENERAL TEST

$$H_0 : C\beta = d, \quad \text{rank}(C) = q$$

Single coefficient

$q=1$; ordinary t test.

Several coefficients

$q>1$; general linear F test.

Nested models

Equivalent to testing whether extra directions are needed.

In rank-deficient models, the rows of C must represent estimable functions.

The general linear F statistic

Standardized squared distance from the null hypothesis.

F TEST

$$F = \frac{(C\hat{\beta} - d)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - d) / q}{MSE}$$

$$F \sim F_{q, n-p} \quad \text{under } H_0$$

Numerator

Signal in the q-dimensional tested subspace.

Denominator

Independent estimate of σ^2 from residual space.

A t test is the one-dimensional case

For one restriction, the F statistic is exactly t^2 .

Q = 1

$$t = \frac{c^T \hat{\beta} - d}{\sqrt{MSE c^T (X^T X)^{-1} c}}$$

$$t \sim t_{n-p} \quad \text{under } H_0$$

$$F_{1,n-p} = t_{n-p}^2$$

Two-sided t test ↔ one-df F test: identical decision.

Global and partial F tests are the same theorem

Only the choice of null subspace changes.

NESTED MODELS

Global regression test

Null model: intercept only.

Extra projection: $H - J$.

$q = p - 1$.

Tests whether all slope directions are unnecessary.

Partial F test

Null model: reduced model X_0 .

Extra projection: $H_1 - H_0$.

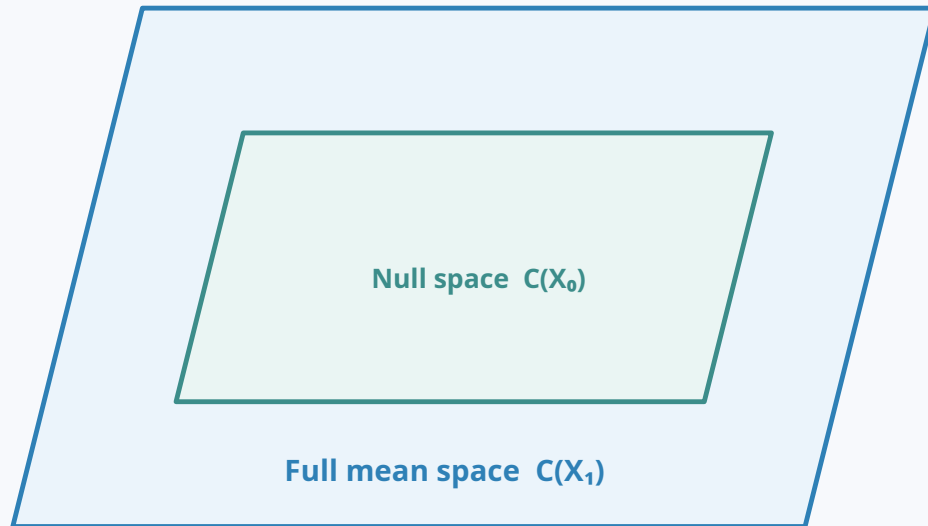
$q = p_1 - p_0$.

Tests whether added directions improve the mean model.

Geometric interpretation of hypothesis testing

A null hypothesis specifies a smaller allowable mean space.

GEOMETRY



$$\text{Evidence against } H_0 \propto \|(H_1 - H_0)Y\|^2$$

Numerator

How far the fitted mean moves outside the null subspace.

Denominator

Noise level estimated in $C(X_1)^\perp$.

Under alternatives: noncentrality measures signal strength

Power increases as the true mean moves farther from the null subspace.

POWER

$$\frac{Y^T P Y}{\sigma^2} \sim \chi_q^2(\lambda), \quad \lambda = \frac{\mu^T P \mu}{\sigma^2}$$

$\lambda = 0$

Null mean has no component in the tested subspace.

$\lambda > 0$

Alternative mean has signal in the tested subspace.

Larger λ

More separation from H_0 and higher power.

Integrated review problem

Work through the chain before looking at the next slide.

PRACTICE

A full-rank model has $n = 30$ and $p = 5$. A reduced nested model has $p_0 = 3$. Suppose $SSE_0 = 140$ and $SSE_1 = 100$.

- What is $\text{rank}(I - H_1)$? What is the distribution of SSE_1/σ^2 ?
- What is $\text{rank}(H_1 - H_0)$? What is the null distribution of $(SSE_0 - SSE_1)/\sigma^2$?
- Why are these two quadratic forms independent?
- Compute the partial F statistic.
- State the numerator and denominator degrees of freedom.

Integrated review problem: solution

Every numerical step corresponds to a projection-space fact.

SOLUTION

Quantity	Result	Reason
$\text{rank}(I - H_1)$	25	$n - p$
SSE_1 / σ^2	χ^2_{25}	projection quadratic form
$\text{rank}(H_1 - H_0)$	2	$p - p_0$
Extra SS / σ^2	χ^2_2 under H_0	null mean annihilated by $H_1 - H_0$
Independence	Yes	$(H_1 - H_0)(I - H_1) = 0$
F	$(40/2)/(100/25) = 5$	ratio of independent mean squares

Minimal formula sheet

Memorize the structural identities; derive the rest.

FORMULA MAP

Topic	Core formula
Estimability	$c \in C(X^T) \Leftrightarrow c \perp N(X)$
Add covariate	$z^* = M_X z; H_+ = H_X + z^* z^{*T} / (z^{*T} z^*)$
MVN transform	$AY + b \sim N(A\mu + b, A\Sigma A^T)$
Projection χ^2	$Z^T P Z \sim \chi^2_{\text{rank}(P)}$
Cochran	orthogonal projections $\Rightarrow$ independent χ^2 pieces
General F	quadratic form in $C\beta - d$ divided by $q \cdot \text{MSE}$

Common midterm mistakes

Most errors come from skipping a structural check.

PITFALLS

Estimability

Testing a nonestimable parameter coordinate in a rank-deficient model.

Noncentrality

Forgetting to check whether $P\mu=0$.

Degrees of freedom

Memorizing df without identifying the underlying rank.

Quadratic forms

Calling every symmetric quadratic form χ^2 .

Independence

Assuming two sums of squares are independent just because they add.

Nested models

Using $H_1 - H_0$ without first establishing $C(X_0) \subset C(X_1)$.

A reliable problem-solving workflow

Use the same sequence on proofs, derivations, and calculations.

EXAM STRATEGY

- 1 Identify the spaces** What are $C(X)$, $N(X)$, and the relevant projection matrices?
- 2 Check rank / estimability** Is the target identified? What is the dimension?
- 3 Express as a transformation** Linear transformation or quadratic form?
- 4 Check orthogonality** Can Cochran or Gaussian zero-covariance independence be used?
- 5 Name the distribution** Normal, χ^2 , t , or F —with the correct df.
- 6 Interpret** What statistical question does the algebra answer?

Midterm review: the unifying picture

- 1 Mean spaces determine what can be estimated.
- 2 Model updating is low-rank projection geometry.
- 3 MVN theory turns linear transformations into exact distributions.
- 4 Projection quadratic forms turn dimensions into chi-squared degrees of freedom.
- 5 Cochran gives the independence needed for exact F tests.
- 6 Hypothesis testing compares nested mean spaces.

Geometry + normality $\Rightarrow$ exact finite-sample inference